Functional Equations

I. Famous Examples

Functional "equations" are equations, system of equations or inequalities whose unknowns are function(s). Famous examples include

1. $f(x+y)=f(x)+f(y)$ (Cauchy)
2. $f(x+y)=f(x)f(y)$ (Cauchy)
3. $f(xy)=f(x)f(y)$ (Cauchy)
4. $f(xy)=f(x)+f(y)$ (Cauchy)
5. $f(\frac{x+y}{2})=\frac{f(x)+f(y)}{2}$ (Jensen)
6. $f(x+y)+f(x-y)=2f(x)f(y)$ (d'Alambert)
7. $g(x+y)=g(x)f(y)+f(x)g(y)$, $f(x+y)=f(x)f(y)-g(x)g(y)$ (trigonometric functions?)
8. $g(x-y)=g(x)f(y)-f(x)g(y)$, $f(x-y)=f(x)f(y)+g(x)g(y)$ (trigonometric functions?)

A functional equation may have many solutions and is in general quite hard to get all of them. By putting more restrictions on the functions (continuous, bounded, monotone, differentiable) one generally gets a smaller set of solutions (or no solution at all).

II. General Methods

There is no systematic way to treat all functional equations. When encountering a functional equation, one may consider: (1) the definition, (2) change of variables to get a suitable form, (3) try solving a system of equations, (4) substitute suitable values into the equation, (5) employ the method of undetermined coefficients, (6) iterative method, (7) find a suitable sequence, (8) use the method of induction, (9) employ suitable parameters, (9) consider fixed points, (10) use the method of contradiction, (11) Cauchy method, etc.

In additions,

- Try the obvious things, obtain obvious solutions, etc.
- Note the domain and range of the function, if the range and domain are natural numbers, then we are basically doing a number theory problem, etc.
- Is the function injective surjective, increasing, decreasing, bounded, periodic, etc.

What are the maximums and minimums of the function?

- Find the fixed points of the functions, what is the smallest fixed point, etc?
- If the unknown is a polynomial, then we can put it into a specific form, etc.

We illustrate by examples.

Example 1 (By definition): Given $f(\frac{1+x}{x})=\frac{x^2+1}{x^2}+\frac{1}{x}$, find $f(x)$.

Indeed $f(\frac{1+x}{x})=(\frac{1+x}{x})^2-\frac{x+1}{x}+1$, hence $f(x)=x^2-x+1$. Because $\frac{1+x}{x}=1+\frac{1}{x}\neq 1$, hence $x=1$ is not in the domain, nor is $x=0$. One verifies the function does work.

Example 2 (By substitution): Given $f(\frac{1+x}{x})=\ln\frac{2x}{1+2x}, x>0$, find $f(x)$.

Let $t=\frac{1+x}{x}$, then $x=\frac{1}{t-1}$, get $f(t)=\ln\frac{2}{t+1}$.

Example 3 (Solve equations): For $x\neq 0,1$, $f(x)+f(\frac{x-1}{x})=1+x$, (1) find $f(x)$. (Putnam)

Replace x by $\frac{x-1}{x}$ in (1), get $f(\frac{x-1}{x})+f(\frac{1}{1-x})=\frac{2x-1}{x}$, (2). Replace x by $\frac{1}{1-x}$ in (1), get $f(\frac{1}{1-x})+f(x)=\frac{2-x}{1-x}$, (3). From (1), (2) and (3), by eliminating $f(\frac{1}{1-x})$ and $f(\frac{x-1}{x})$, get $f(x)=\frac{x^3-x^2-1}{2x(x-1)}$.

Example 4 (By putting in suitable values): $f:\mathbb{R}\to\mathbb{R}$ and is not identically zero, $f\left(\frac{\pi}{2}\right)=0$, and for all $x,y\in\mathbb{R}$, $f(x)+f(y)=2f(\frac{x+y}{2})f(\frac{x-y}{2})$ (*). Show that (1) *f* is periodic of 2π, (2) $f(x)=f(-x)$ (even), and (3) $f(2x)=2f^2(x)-1$.

Replace x, y in (*) by $x+\pi, x$, get $f(x+\pi)+f(x)=2f(x+\frac{\pi}{2})f(\frac{\pi}{2})=0$, hence $f(x+\pi)=-f(x)$. Therefore $f(x+2\pi)=f(x+\pi+\pi)=-f(x+\pi)=f(x)$, f is periodic of period 2π. Now if $x=y$ in (*), then $2f(x)=2f(x)f(0)$. As *f* is not identically zero, we can choose x so that $f(x) \neq 0$, get $f(0)=1$. Thus by (*) again, get $f(x)+f(-x)=2f(0)f(x)=2f(x)$, hence $f(x)=f(-x)$ (even). Finally replace x, y by $2x, 0$, get $f(2x)=2f^2(x)-1$.

Example 5 (Undetermined coefficients) Given $f(x)$ a polynomial and such that $f(x+1)+f(x-1)=2x^2-2x+4$, find $f(x)$.

Clearly $f(x)$ is a polynomial of degree 2, so let $f(x)=ax^2+bx+c$. Put into the equation and simplify to get $f(x)=x^2-x+1$.

Example 6 (Iterative method) Let $f:\mathbb{N}\to\mathbb{N}$ and such that for all $m,n\in\mathbb{N}$, $f(m+n)=f(m)+f(n)+mn$, $f(1)=1$. Find $f(x)$.
Let $m=1$, get $f(n+1)=f(1)+f(n)+n=f(n)+n+1$. Then replace n by $1,2,\dots,n-1$, get $f(2)=f(1)+2$, $f(3)=f(2)+3$, $\dots$, $f(n)=f(n-1)+n$. Summing up the equations to get $f(n)=\dfrac{n(n+1)}{2}$.

Example 7 (Sequence): Let $f:\mathbb{N}\to\mathbb{N}$, $f(1)=a$ and such that $f(n)=pf(n-1)+q$, where $p\neq 1$, q are constants. Find $f(n)$.
Check $\dfrac{f(n+1)-f(n)}{f(n)-f(n-1)}=p$, hence $\{f(n+1)-f(n)\}$ a geometric sequence of ratio p, get $f(n+1)-f(n)=((f(2)-f(1))p^{n-1}$. Enter other data to get
$f(n)=(a+\dfrac{q}{p-1})p^{n-1}-\dfrac{q}{p-1}$.

Example 8 (Induction) Let $f:\mathbb{N}\to\mathbb{N}^{\geq 0}$ be such that (i) $f(m+n)-f(m)-f(n)=0$ or 1, for all $m,n\in\mathbb{N}$, (ii) $f(2)=0, f(3)>0$ and $f(9999)=3333$. Determine $f(1982)$. (IMO 1982)
Put $m=n=1$, get $f(2)-f(1)-f(1)\geq 0$, but $f(2)=0$, get $f(1)=0$. Put $m=2, n=1$, get $f(3)=0$ or 1, hence $f(3)=1$. Then $f(3+3)\geq f(3)+f(3)=2$. An easy induction shows that $f(3n)\geq n$.
$(f(3(k-1)+3)\geq f(3(k-1))+f(3)\geq k-1+1=k.)$
Moreover if $f(3n)>n$ for some n, then $f(3m)>m$, for all $m\geq n$. Since $f(9999)=3333$, we must have $f(3n)=n$, for all $n\leq 3333$. Thus
$1982=f(3\cdot 1982)\geq f(2\cdot 1982)+f(1982)\geq 3f(1982)$. Therefore
$f(1982)\leq\dfrac{1982}{3}<661$. Also
$f(1982)\geq f(1980)+f(2)=f(3\cdot 660)=660$. We get $f(1982)=660$.

Example 9 (Iterative Functions) Let $k\in\mathbb{N}$ and $f:\mathbb{N}\to\mathbb{N}$ is a strictly increasing function such that $f(f(n)) = kn$ for every $n \in \mathbb{N}$. Show that

$\frac{2k}{k+1} n \le f(n) \le \frac{k+1}{2} n$ for every $n \in \mathbb{N}$. (CMO 1990)

As f is strictly increasing

$f(n) \ge f(n-1)+1 \ge f(n-2)+2 \ge \ldots \ge f(1)+(n-1) \ge n$ (1).

Now let $f(n) = a$, say, then $a \ge n$, and

$k^2 n = f(ka) \ge f(ka-1)+1 \ge \ldots \ge f(kn)+ka-kn = 2ka-kn$, which implies

$a = f(n) \le \frac{k+1}{2} n$. On the other hand, from (1),

$f(f(n)) \ge f(n)$, or $kn \ge a$. Hence

$ka = kf(n) \ge f(kn-1)+1 \ge \ldots \ge f(a)+kn-a = 2kn-a$, which implies

$a = f(n) \ge \frac{2k}{k+1} n$. Together the result follows.

<u>Example 10</u> (Fixed Points) Find all $f : \mathbb{R}^{>0} \to \mathbb{R}^{>0}$ such that for all $x, y \in \mathbb{R}^{>0}$, $f(xf(y)) = yf(x)$ and $f(x) \to 0$ when $x \to \infty$. (IMO1983)

Let $a, x_0 \in \mathbb{R}^{>0}$ be arbitrary positive real numbers. As $f(x_0) > 0$, hence $y_0 = \frac{a}{f(x_0)} > 0$. This implies $f(x_0 f(y_0)) = y_0 f(x_0) = a$, hence any positive real number is in the range of f. In particular we can find y such that $f(y) = 1$. Now $f(1) = f(1f(y)) = yf(1)$, which implies $y = 1$. Hence $y = 1$ is a fixed point of f. Moreover as $f(xf(x)) = xf(x)$, hence for any $x > 0$, $xf(x)$ is a fixed point of f. For fixed points a and b, i.e. $f(a) = a$ and $f(b) = b$, we have $f(ab) = f(af(b)) = bf(a) = ba$, hence ab is also a fixed point. This implies for any fixed point a, so is a^k for any positive integer k. If $a > 1$ a fixed point, then $f(a^k) = a^k \to \infty$ as $k \to \infty$, violating the condition $f(x) \to 0$. Therefore $a \le 1$. Or for any x, $xf(x) \le 1$. On the other hand, for any $x > 0$,

$1 = f(1) = f(\frac{1}{xf(x)} f(xf(x))) = xf(x) f(\frac{1}{xf(x)})$, hence $\frac{1}{xf(x)}$ is also a fixed point. By the previous argument, we have also $\frac{1}{xf(x)} \le 1$. Together we must have $f(x) = \frac{1}{x}$ for all x.

<u>Example 11</u> (By contradiction): Let $f : \mathbb{N} \to \mathbb{N}$ be a strictly increasing function such that $f(2) = 2$, and $f(mn) = f(m)f(n)$ when m and n are relatively prime. Show that $f(n) = n$ for all n. (Putnam)

Indeed $2 = f(2) = f(2)f(1) = 2f(1)$, thus $f(1) = 1$. Also
$f(3)f(7) = f(21) < f(22) = f(2)f(11) = 2f(11) < 2f(14) = 2f(2)f(7) = 4f(7)$,
hence $f(3) < 4$. But $f(3) > f(2) = 2$, we must have $f(3) = 3$.
Now we assume the statement is false, there exist the *smallest positive integer* $n_0 \geq 4$, such that $f(n_0) \neq n_0$. This means $f(1) = 1, f(2) = 2, \ldots, f(n_0 - 1) = n_0 - 1$, but $f(n_0) \neq n_0$. Therefore we must have $f(n_0) > n_0$. Consequently $f(n) > n$ for all $n \geq n_0$ (*). Now if n_0 is odd, then 2 and $n_0 - 2$ are relatively prime, hence
$f(2(n_0 - 2)) = f(2)f(n_0 - 2) = 2(n_0 - 2) \geq n_0$, contradicting (*). If n_0 is even, then 2 and $n_0 - 1$ are relatively prime, and again
$f(2(n_0 - 1)) = f(2)f(n_0 - 1) = 2(n_0 - 1) \geq n_0$, contradicting (*). Together we observe such a n_0 does not exist.

<u>Example 12</u> (Cauchy Method): $f : \mathbb{R} \to \mathbb{R}$ be continuous and such that for any $x, y \in \mathbb{R}$, $f(x + y) = f(x) + f(y)$. Find $f(x)$.
(The method means start with cases for integers and rational numbers and then to all real numbers.) We give the outlines. First show by induction that
$f(x_1 + x_2 + \ldots + x_n) = f(x_1) + f(x_2) + \ldots f(x_n)$. Let $x = x_1 = x_2 = \ldots = x_n$, get $f(nx) = nf(x)$. Let $x = 1$ and $f(1) = a$, get $f(n) = an$ for all positive integers *n*.
Put $y = 0$, get $f(0) = 0$. Hence $0 = f(0) = f(n - n) = f(n) + f(-n)$, get $f(-n) = -f(n) = -an = a(-n)$, hence the formula $f(n) = n$ is true for all integers *n*.
Now $am = f(m) = f(n\frac{m}{n}) = nf(\frac{m}{n})$, hence $f(\frac{m}{n}) = a\frac{m}{n}$, the formula is valid for rational numbers. By taking limits to reals and using the fact *f* is continuous to get the required result $f(x) = ax$ for all $x \in \mathbb{R}$.
(The statement is not true if continuity not assumed (G. Hamel, 1905)).

<u>Example 13</u> $f(x + y) = f(x)f(y)$.
If $f(a) = 0$, then $f(x + a) = f(x)f(a) = 0$ for all *x*, hence $f(x) \equiv 0$. For all other solutions $f(x) \neq 0$ everywhere. Now put $x = y = \frac{t}{2}$, get $f(t) = f^2(\frac{t}{2}) > 0$. The solutions should be everywhere positive. Now $f(x + 0) = f(x)f(0)$, get $f(0) = 1$. Now by induction we can show $f(nx) = f^n(x)$. By the same token we have $f(\frac{n}{m}x) = f^{\frac{n}{m}}(x)$. Put $x = 1$ and let $f(1) = a$, we get $f(\frac{n}{m}) = a^{\frac{n}{m}}$. By continuity argument we get $f(x) = a^x$ for all real *x*.

III. Miscellaneous Examples

Example 14: Show that there doesn't exist a function *f*, $f:\mathbb{N}_0 \to \mathbb{N}_0$, show that $f(f(n)) = n+1987$. (IMO1987)

(Note: $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$.)

If so, we have $f(f(f(n))) = f(n+1987) = f(n)+1987$, for all $n \geq 0$. Hence $f(n+1987q) = 1987q + f(n)$ by iterations. So such a function is defined by its range on the set $S = \{0,1,2,...,1986\}$. Now let $r \in S$, and suppose $f(r) = 1987k + r_1$, with $r_1 \in S$. Then $1987+1987 > r+1987 = f(f(r)) = f(1987k + r_1) = 1987k + f(r_1)$, we get $k=0$ or 1. When $k=0$, we have $f(r) = r_1 \leq 1986$ and $f(r_1) = r+1987$. When $k=1$, we get $f(r_1) = r$. In this way we obtain a function $g: S \to S$, defined by $g(r) = r_1$, r_1 the residue of $f(r)$ mod 1987. We see that $g \circ g = 1_S$ (order = 2) and *g* has no fixed points. Then *g* is a product of distinct transpositions, $g = t_1 t_2 ... t_p$, with $2p = |S|$, contradicting $|S| = 1987$. (Alternate: because $f(r) = r_1$ and $f(r_1) = r$, we can pair up the pairs and there must exist a fixed point $f(n) = n + 1987k$, with $n \in S$. Hence

$n+1987 = f(f(n)) = f(n+1987k) = f(n)+1987k$, thus $n+1987 = f(n)+1987k$. Therefore $f(n) = n - 1987(k-1)$. Solve to get $1987k = -1987(k-1)$, or $2k = 1$, a contradiction.)

Example 15 Find all $f:\mathbb{R}^{\geq 0} \to \mathbb{R}^{\geq 0}$ such that (i) $f(xf(y))f(y) = f(x+y)$ for all $x, y \geq 0$, (ii) $f(2) = 0$, and (iii) $f(x) \neq 0$ for $0 \leq x < 2$. (IMO 1986)

Put $y=2$ in (i), get $f(xf(2))f(2) = f(x+2) = 0$ for all $x \geq 0$, hence $f(x) = 0$ when $x \geq 2$. Now let $x \geq 0$ and $0 \leq y < 2$ such that $x+y \geq 2$. Use (i) and (iii), get $0 = f(x+y) = f(xf(y))f(y)$, hence $xf(y) \geq 2$, or $x \geq \dfrac{2}{f(y)}$. Reciprocally $xf(y) \geq 2$, then $f(x+y) = 0$ or $x+y \geq 2$. Thus for any $0 \leq y < 2$ and any $x \geq 0$, the conditions $x \geq 2-y$ and $x \geq \dfrac{2}{f(y)}$ are equivalent. Consequently $f(y) = \dfrac{2}{2-y}$ when $0 \leq y < 2$ and $f(y) = 0$ when $y \geq 2$. One can verify that the function does satisfy the requirements.

Example 16 Construct a function $f:\mathbb{Q}^{>0} \to \mathbb{Q}^{>0}$ such that $f(xf(y)) = \dfrac{f(x)}{y}$ for all $x, y \in \mathbb{Q}^{>0}$. (IMO1990)

First show that f is injective. Indeed if $f(y_1)=f(y_2)$, then $\dfrac{f(x)}{y_1}=\dfrac{f(x)}{y_2}$, for all x, hence $y_1=y_2$. Put $y=1$, get $f(xf(1))=\dfrac{f(x)}{1}=f(x)$, hence $xf(1)=x$ for all x, get $f(1)=1$. Put $x=1$, get $f(f(y))=\dfrac{1}{y}$ for all y. Now apply f again, get $f(\dfrac{1}{y})=\dfrac{1}{f(y)}$. Finally set $y=f(\dfrac{1}{t})$, get $f(xt)=f(xf(f(\dfrac{1}{t})))=\dfrac{f(x)}{f(1/t)}=f(x)f(t)$, for all $x,t\in\mathbb{Q}^{>0}$.

Conversely it is easy to check that if $f(xt)=f(x)f(t)$ and $f(f(x))=\dfrac{1}{x}$, then f solves the equations.

Denote by p_i the i^{th} prime, we may define f by $f(p_i)=\begin{cases} p_{i+1}, & i \text{ odd} \\ \dfrac{1}{p_{i-1}}, & i \text{ even.}\end{cases}$ Then we extend f by defining $f(p_1^{k_1}p_2^{k_2}\ldots p_n^{k_n})=f^{k_1}(p_1)f^{k_2}(p_2)\ldots f^{k_n}(p_n)$ to obtain a feasible solution.

<u>Example 17</u>: Find all $f:\mathbb{N}_0\to\mathbb{N}_0$ such that $f(m+f(n))=f(f(m))+f(n)$ for all $m,n\in\mathbb{N}_0$. (IMO 1996)

Set $m=n=0$, get $f(f(0))=f(f(0))+f(0)$, and hence $f(0)=0$. Also if we set $m=0$, we get $f(f(n))=f(n)$, for all n. Therefore one may reduce the given conditions to $f(0)=0$ and $f(m+f(n))=f(m)+f(n)$. Clearly $f(x)\equiv 0$ is a solution. We look for other non-trivial solutions. Let a be the *smallest fixed point* of f. If $a=1$, then one shows by induction $f(n)=n$ for all n. So now we assume $a>1$. By induction all ka are fixed points. If b is another fixed point, then $b=aq+r$, (Euclidean algorithm). Then
$b=f(b)=f(aq+r)=f(r+f(qa))=f(r)+qa$, hence $f(r)=r$, or r a fixed point, forcing $r=0$. Thus all fixed points are of the form ka. Now for any $i, 0<i\le a-1$, as $f(i)$ is a fixed point, we must have $f(i)=n_i a$, where $n_i\in\mathbb{N}_0$ and $n_0=0$. If $n=aq+i$, we extend f by the formula
$f(n)=f(aq+i)=f(aq)+f(i)=qa+n_i a=(\left\lfloor\dfrac{n}{a}\right\rfloor+n_i)a$. One verifies the formula does satisfy the requirement.

IV. Exercises

1. The set of all positive integers is the union of two disjoint subsets $\{f(1), f(2), ..., f(n), ...\}$ and $\{g(1), g(2), ..., g(n), ...\}$ where $f(1) < f(2) < ...$, $g(1) < g(2) < ...$ and $g(n) = f(f(n)) + 1$, for all $n \geq 1$. Determine $f(240)$. (IMO 1978)
2. The function $f(x, y)$ satisfies the conditions: (i) $f(0, y) = y + 1$, (ii) $f(x+1, 0) = f(x, 1)$ and (iii) $f(x+1, y+1) = f(x, f(x+1, y))$ for all non-negative integers *x* and *y*. Determine $f(4, 1981)$. (IMO 1981)
3. Let $f : \mathbb{N} \to \mathbb{N}$ be a function satisfying $f(1) = 1, f(3) = 3$, $f(2n) = f(n)$, $f(4n+1) = 2f(2n+1) - f(n)$ and finally $f(4n+3) = 2f(2n+1) - 2f(n)$ for all positive integers *n*. Determine the number of positive integers *n*, $n \leq 1988$, for which $f(n) = n$. (IMO 1988)
4. Determine all $f : \mathbb{R} \to \mathbb{R}$ such that $f(x^2 + f(y)) = y + (f(x))^2$ for all $x, y \in \mathbb{R}$. (IMO 1992)
5. Determine all $f : (-1, \infty) \to (-1, \infty)$ satisfying (i) $f(x + f(y) + xf(y)) = y + f(x) + yf(x)$ for all $x, y \in (-1, \infty)$, and (ii) $\dfrac{f(x)}{x}$ is strictly increasing on the intervals $(-1, 0)$ and $(0, \infty)$. (IMO 1994)
6. Consider all functions $f : \mathbb{N} \to \mathbb{N}$ satisfying $f(t^2 f(s)) = s(f(t))^2$, for all $s, t \in \mathbb{N}$. Determine the least possible value of $f(1998)$. (IMO 1998)
7. Determine all $f : \mathbb{R} \to \mathbb{R}$ such that $f(x - f(y)) = f(f(y)) + xf(y) + f(x) - 1$ for all $x, y \in \mathbb{R}$. (IMO1999)
8. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $(f(x) + f(z))(f(y) + f(t)) = f(xy - zt) + f(xt + yz)$ for all x, y, z, t in $\mathbb{R}$. (IMO 2002)
9. Find all functions $f : (0, \infty) \to (0, \infty)$ such that $\dfrac{f(p)^2 + f(q)^2}{f(r^2) + f(s^2)} = \dfrac{p^2 + q^2}{r^2 + s^2}$ for all $p, q, r, s > 0$ with $pq = rs$. (IMO 2008)
10. Determine all functions $f : \mathbb{N} \to \mathbb{N}$ such that for all *x* and *y* there exists a non-degenerate triangle of sides $x, f(y)$ and $f(y + f(x) - 1)$. (IMO 2009)

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1. The set of all positive integers is the union of two disjoint subsets $\{f(1), f(2), ..., f(n), ...\}$ and $\{g(1), g(2), ..., g(n), ...\}$ where $f(1) < f(2) < ...,$ $g(1) < g(2) < ...$ and $g(n) = f(f(n)) + 1,$ for all $n \geq 1$. Determine $f(240)$. (IMO 1978)

$f(240) = 388.$ Try to fill the gaps to get the answer. This problem clearly has background of higher mathematics.

2. The function $f(x, y)$ satisfies the conditions: (i) $f(0, y) = y + 1,$ (ii) $f(x+1, 0) = f(x, 1)$ and (iii) $f(x+1, y+1) = f(x, f(x+1, y))$ for all non-negative integers *x* and *y*. Determine $f(4, 1981)$. (IMO 1981)

Using a lot of inductions show that $f(1, y) = y + 2,$ $f(2, y) = 2y + 3,$

$f(3, y) = 2^{y+3} - 3$ and finally $f(4, y) + 3 = 2^{2^{\cdot^{\cdot^{2}}}}$ ($y + 3$ two's). Hence

$f(4, 1981) = 2^{2^{\cdot^{\cdot^{2}}}} - 3$ (1984 two's).

3. Let $f : \mathbb{N} \to \mathbb{N}$ be a function satisfying $f(1) = 1, f(3) = 3,$ $f(2n) = f(n),$ $f(4n+1) = 2f(2n+1) - f(n)$ and finally $f(4n+3) = 2f(2n+1) - 2f(n)$ for all positive integers *n*. Determine the number of positive integers *n*, $n \leq 1988,$ for which $f(n) = n.$ (IMO 1988)

When writing *n* as a binary number, *f* essentially swaps the head and tail of the numbers. Hence we look for palindromes less than or equal to 1988, get 92 of them.

4. Determine all $f : \mathbb{R} \to \mathbb{R}$ such that $f(x^2 + f(y)) = y + (f(x))^2$ for all $x, y \in \mathbb{R}.$ (IMO 1992)

$f(x) = x$ is the only solution.

5. Determine all $f : (-1, \infty) \to (-1, \infty)$ satisfying (i) $f(x + f(y) + xf(y)) = y + f(x) + yf(x)$ for all $x, y \in (-1, \infty),$ and (ii) $\dfrac{f(x)}{x}$ is strictly increasing on the intervals $(-1, 0)$ and $(0, \infty).$ (IMO 1994)

$$f(x) = -\frac{x}{1+x}.$$

6. Consider all functions $f:\mathbb{N}\to\mathbb{N}$ satisfying $f(t^2 f(s)) = s(f(t))^2$, for all $s,t\in\mathbb{N}$. Determine the least possible value of $f(1998)$. (IMO 1998)

First let $f(1)=a$. Then show $af(mn)=f(m)f(n)$. Then shows $f(n)$ is divisible by a for all n. Let $g(n)=\dfrac{f(n)}{a}$, then g satisfies $g(1)=1, g(mn)=g(m)g(n)$ and $g(g(n))=n$. Now show that g sends distinct primes to distinct primes. As $1998=2\times 3^3\times 37$, we can define g by sending 2 to 3, 3 to 2, 5 to 37, 37 to 5 and all other p to p, then extend g using multiplicity. Then $g(1998)=2^3\times 3\times 5=120$.

7. Determine all $f:\mathbb{R}\to\mathbb{R}$ such that $f(x-f(y))=f(f(y))+xf(y)+f(x)-1$ for all $x,y\in\mathbb{R}$. (IMO1999)

$f(x)=1-\dfrac{x^2}{2}$.

8. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that $(f(x)+f(z))(f(y)+f(t))=f(xy-zt)+f(xt+yz)$ for all x,y,z,t in $\mathbb{R}$. (IMO 2002)

The equation has the solution $f(x)\equiv 0$, for all x, $f(x)\equiv\dfrac{1}{2}$ for all x and $f(x)=x^2$ for all x. These make both sides of the equation equal 0, 1 and $(x^2+z^2)(y^2+t^2)$ respectively, and there are no other solutions. Put $x=u=z=t=0$, get $4f^2(0)=2f(0)$. Hence $f(0)=0$ or $\dfrac{1}{2}$. If $f(0)=\dfrac{1}{2}$, we have $\dfrac{1}{2}+f(t)=(f(0)+f(0))(f(0)+f(t))=f(0)+f(0)=1$, for all t, hence $f(t)\equiv\dfrac{1}{2}$. If $f(0)=0$, then $(f(x)+f(0))(f(y)+f(0))=f(xy)+f(0)$, hence $f(xy)=f(x)f(y)$, thus f is multiplicative, so $f(1)=f^2(1)$, thus $f(1)=0$ or 1. If $f(1)=0$, then $f(x)=f(x)f(1)=0$, hence $f(x)\equiv 0$. Now if $f(0)=0$ and $f(1)=1$, then by putting $x=0$ and $y=t=1$ into the original equation, get $f(-z)+f(z)=2f(z)$, or $f(z)=f(-z)$ and f is even. So it suffices to consider positive x. Now note that $f(x^2)=f^2(x)\ge 0$, as f is even and x positive, we get $f(y)\ge 0$ for all y. Now put $t=x$ and $z=y$, get $f(x^2+y^2)=(f(x)+f(y))^2$. This shows $f(x^2+y^2)\ge f(x^2)=f^2(x)$. Hence f is increasing on the positive side. Finally set $y=z=t=1$, get $f(x-1)+f(x+1)=2(f(x)+1)$. Using this, and by induction on n, get $f(n)=n^2$ for all natural numbers n and hence for all integers n.

As f is multiplicative, $f(a)=a^2$ for all rational a. Now suppose $f(x)\neq x^2$ for some x. If $f(x)<x^2$, take a rational a with $x>a>\sqrt{f(x)}$, then $f(a)=a^2>f(x)$, contradicts the fact that f is increasing. Likewise $f(x)>x^2$ is impossible.

9. Find all functions $f:(0,\infty)\to(0,\infty)$ such that $\frac{f(p)^2+f(q)^2}{f(r^2)+f(s^2)}=\frac{p^2+q^2}{r^2+s^2}$ for all $p,q,r,s>0$ with $pq=rs$. (IMO 2008)

Check that $f(x)=x$ or $f(x)=\frac{1}{x}$ are the only solutions.

10. Determine all functions $f:\mathbb{N}\to\mathbb{N}$ such that for all x and y there exists a non-degenerate triangle of sides $x, f(y)$ and $f(y+f(x)-1)$. (IMO 2009)

First show $f(1)=1$. Indeed if $f(1)=1+m>1$, then $f(y)=f(y+m)$ for all y considering the triangle with the side lengths 1, $f(y)$ and $f(y+m)$. Thus f is m-periodic and hence bounded, say by B. Then we choose $x>2B$ to obtain a contradiction. Second step: Set $x=z$ and $y=1$, get $f(f(z))=z$. Step 3: Show by contradiction $f(z)\leq z$. Finally $z=f(f(z)\leq f(z)\leq z$, the result follows.

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